Data experiments: source value, mixture weights, and repetition

> **Reading note.** This is an explanatory reconstruction of Jiaxuan Zou's literature synthesis. Published experiments belong to the cited teams. Numbered protocols and illustrative matrices are our planning synthesis unless explicitly attributed.

The blog's §7 contains three distinct experiments: screening a new source, optimizing domain weights, and measuring the value of repeated data. Hold architecture and optimization fixed initially so that a source's effect can be identified.

Inventory and source screening

1. Deduplicate across sources, record provenance, and freeze quality filters and tokenizer.
2. For bucket j, count unique tokens U_j after filtering and deduplication. Version the resulting manifest.
3. Screen a source using a controlled small-model addition or a short continuation branch from a common checkpoint.
4. Score every candidate on identical validation distributions and domain tasks. A micro-annealing result measures marginal value at that checkpoint, not a full-training ranking.

OLMo 2 uses short continuation experiments in its data-development process; the blog combines this with the mixture methodologies of Olmix, OLMo 3, and Marin. These methods have different experimental scopes and should not be treated as interchangeable evidence.

Fit a constrained mixture

Let $w_j \geq 0$ be a bucket's fraction of D loss-bearing training tokens, with $\sum_j w_j = 1$. Its approximate average exposure is

$$e_j = \frac{D w_j}{U_j}.$$

An epoch cap k_j implies $w_j \leq k_j U_j / D$. This accounting assumes the declared within-bucket sampling process and does not capture document-level exposure variation.

1. Establish token-proportional and repetition-capped baselines.
2. Sample candidate weights over the feasible simplex; use a declared prior.
3. Train proxies with matched optimizer and horizon conventions. Downscale bucket pools to reproduce target exposure when studying repetition.
4. Fit per-task response models and validate them on mixtures excluded from fitting. Examine non-monotonic saturation.
5. Optimize the declared aggregate objective subject to caps. If using KL regularization toward a prior, tune its coefficient on development mixtures.
6. Train the proposed mixture at a larger scale and compare it against the baseline with paired seeds.

For repetition, one model in Muennighoff et al. uses

$$D_{\text{eff}} = U \left[1 + R_* \left(1 - e^{-R/R_*} \right) \right],$$

where $R = D/U - 1 \geq 0$ is additional repetition beyond the first epoch and $R_* > 0$ is a fitted decay scale. R_* is an e-folding scale; the marginal contribution halves after $R_* \ln 2$, so it should not be called a half-life. The expression describes diminishing returns rather than all possible overfitting dynamics.

The cited paper's Appendix A reports $R_D^* = 15.387756$ and $R_N^* = 5.309743$ for its extended fit. These are recipe-specific fitted values. The supplied notes correctly distinguish the roughly four-epoch near-fresh regime from the approximately 15.4 data-repetition decay scale. Muennighoff et al., v5, Eq. 14.

Minimal matrix and acceptance

Compare natural weights, capped weights, sampled mixtures, and the solved mixture at a fixed proxy size/horizon; confirm finalists at an intermediate scale. Sweep unique-pool size or epochs separately to identify repetition effects.

Acceptance checklist: fixed validation; audited overlaps; feasible epoch caps; held-out mixture predictions; intermediate-scale confirmation; no regression in critical domains; combined optimizer/mixture check before production.

Sources: blog §7; OLMo 2, 2025; Olmix, 2026; OLMo 3, 2025, §3.4; Marin data pipeline; Muennighoff et al., 2023.

Blog/notes crosswalk · Detailed companion